

What is Machine Learning?

Machine Learning (ML) is a part of Artificial Intelligence where computers learn from data instead of being explicitly programmed.

Simple definition:

“Machine learning is teaching computers to learn patterns from data and make decisions.”

Real-life examples:

- YouTube recommending videos
- Netflix suggesting movies
- Google Maps predicting traffic
- Spam email detection

Example:

Instead of writing rules like:

```
if email contains "win money" → spam
```

Machine Learning learns automatically from thousands of emails.

Why Should You Learn Machine Learning?

Since you want to build a career in the coding field, ML is a very powerful skill.

1. High Demand Career

- Data Scientist
- ML Engineer
- AI Engineer

These roles are highly paid.

2. Real-World Impact

ML is used in:

- Healthcare (disease prediction)
- Finance (fraud detection)
- E-commerce (recommendations)

3. Future Technology

ML is the backbone of:

- Chatbots (like ChatGPT)
- Self-driving cars
- Face recognition

4. Smart Problem Solving

Instead of writing complex logic, ML:
Learns patterns automatically

What is Scikit-learn?

Scikit-learn is a Python library used to build machine learning models easily.

It is one of the most popular ML libraries for beginners.

What can you do with Scikit-learn?

- Classification (Spam detection)
- Regression (Price prediction)
- Clustering (Customer grouping)
- Data preprocessing

Example (Very Simple)

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()

X = [[1], [2], [3]]
y = [2, 4, 6]

model.fit(X, y)

print(model.predict([[4]])) # Output: 8
```

Here the model learned:

$$y = 2x$$

How Much Maths is Required for Machine Learning?

Beginner level (enough to start ML):

- Basic Algebra
- Basic Statistics
- Basic Probability
- Very basic Linear Algebra

Good news:

You do NOT need advanced mathematics (like heavy calculus proofs) in the beginning.

Maths Concepts You Should Learn (Step-by-Step)

1. Algebra (Most Important)

- Variables (x, y)
- Equations
- Linear equations
- Graph understanding

ML models learn relationships like these.

2. Basic Statistics

- Mean (average)
- Median
- Mode
- Variance
- Standard Deviation

Used to understand data in ML.

3. Probability (Important for ML logic)

- Probability basics
- Conditional probability
- Independent events

Used in predictions and spam detection.

4. Basic Linear Algebra

- Vectors (list of numbers)
- Matrices (table of numbers)

Data in ML is often represented in matrix form.

Practice Questions (with Answers)

If you can solve these → You are ready to start ML 🚀

Algebra Questions

Q1:

If $y = 2x + 3$, find y when $x = 5$

Answer:

$$y = 2(5) + 3 = 13$$

Q2:

Solve: $2x = 10$

Answer:

$$x = 5$$

Statistics Questions

Q3:

Find mean of: 2, 4, 6, 8

Answer:

$$\text{Mean} = (2 + 4 + 6 + 8) / 4 = 5$$

Q4:

Find median of: 1, 3, 5, 7, 9

Answer:

$$\text{Median} = 5$$

Probability Questions

Q5:

What is the probability of getting heads in a coin toss?

Answer:

$$1/2 = 0.5$$

Q6:

A bag has 3 red and 2 blue balls.

What is the probability of picking a red ball?

Answer:

$$3 / 5 = 0.6$$

Linear Algebra Questions

Q7:

Vector addition:

$$[1, 2] + [3, 4]$$

Answer:

$$[4, 6]$$

Q8:

Find the element in the 1st row and 2nd column of the matrix:

$$\begin{bmatrix} 1 & 5 \\ 2 & 8 \end{bmatrix}$$

Answer:

5

Functions (Very Important for ML)

A **function** is a relationship between input and output.

Example:

If:

- $f(2) = 4$
- $f(3) = 9$

ML models = functions that map input $\rightarrow$ output

Questions

Q1:

If $f(x) = x^2$, find $f(4)$

Answer:

$$f(4) = 16$$

Q2:

If $f(x) = 3x + 2$, find $f(5)$

Answer:

$$= 15 + 2 = 17$$

Graph Understanding (Visualization)

Graphs show how input (x) affects output (y)

Example:

Shape: U-shaped curve

ML uses graphs to:

- understand trends
- visualize predictions

Questions

Q3:

For $y = x^2$, what is y when $x = -2$?

Answer:

$$(-2)^2 = 4$$

Q4:

Is $y = x^2$ linear or non-linear?

Answer:

Non-linear

Loss Function (Core ML Concept)

Concept:

Loss function tells:

"How wrong your model is"

Example (Mean Squared Error):

y = actual value

$\hat{y}$ = predicted value

Questions

Q5:

Actual = 5, Predicted = 3

Find error = $(y - \hat{y})$

Answer:

$$= 5 - 3 = 2$$

Q6:

Square error?

Answer:

$$2^2 = 4$$

Gradient (Basic Idea Only)

Gradient = **slope of a function**

Example:

$$\text{Slope} = 2$$

ML uses gradient to:

- reduce error
- improve model

Questions

Q7:

What is slope of $y = 5x$?

Answer:

5

Q8:

If slope is positive, graph goes?

Answer:

Upward

Distance (Used in KNN Algorithm)

Concept:

Distance helps measure similarity

Formula:

Questions

Q9:

Find distance between (1,2) and (4,6)

Answer:

$$= \sqrt{[(4-1)^2 + (6-2)^2]}$$

$$= \sqrt{9 + 16}$$

$$= \sqrt{25} = 5$$

10. Feature & Target (ML Terminology)

- **Feature (X)** → input
- **Target (y)** → output

Example:

Hours Study	Marks
2	30

Feature = Hours

Target = Marks

Questions

Q10:

In house price prediction:

- Size of house = ?

Answer:

Feature

Q11:

Price of house = ?

Answer:

Target

Accuracy, Precision, Recall (ML Evaluation Metrics)

These are **metrics used to evaluate the performance of a machine learning model**, especially in classification problems.

Used widely with libraries like Scikit-learn.

1. Accuracy

Accuracy = Number of correct predictions out of total predictions

Formula:

$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$

Meaning:

“How often the model is correct overall”

2. Precision

Precision = Out of predicted positive cases, how many are actually positive

Formula:

$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$

Where:

- TP = True Positive
- FP = False Positive

Meaning:

“When the model predicts YES, how often is it correct?”

3. Recall

Recall = Out of actual positive cases, how many were correctly predicted

Formula:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Where:

- FN = False Negative

Meaning:

“Out of all real YES cases, how many did the model catch?”

4. Biotech Example (Disease Detection)

Suppose you are building a model to detect a disease (e.g., cancer).

Dataset:

- Total patients = 100
- Actually diseased = 20
- Healthy = 80

Model Predictions:

- True Positive (TP) = 15 → correctly detected disease
- False Positive (FP) = 10 → wrongly predicted disease
- False Negative (FN) = 5 → missed disease
- True Negative (TN) = 70 → correctly predicted healthy

Calculations

1. Accuracy:

$$\text{Correct predictions} = \text{TP} + \text{TN} = 15 + 70 = 85$$

$$\text{Accuracy} = 85 / 100 = \mathbf{85\%}$$

2. Precision:

$$\begin{aligned} \text{Precision} &= 15 / (15 + 10) \\ &= 15 / 25 = \mathbf{60\%} \end{aligned}$$

3. Recall:

$$\begin{aligned} \text{Recall} &= 15 / (15 + 5) \\ &= 15 / 20 = \mathbf{75\%} \end{aligned}$$

4. Interpretation

Accuracy = 85%

The model looks good overall

Precision = 60%

Problem:

- 40% of predicted disease cases are wrong
(Too many false alarms)

Recall = 75%

Good:

- Model detects 75% of actual patients
(Still misses 25% — risky)

Types of Machine Learning

There are **3 main types**:

1. Supervised Learning
2. Unsupervised Learning
3. Reinforcement Learning

1. Supervised Learning

Supervised Learning is a type of machine learning where the model is trained using **labeled data**.

Labeled data = Input + Correct Output already given

Simple Idea:

“Learning with a teacher”

You already know the correct answers, and the model learns from them.

Example (Biotech):

Disease prediction:

Symptoms	Disease
Fever, cough	Flu
Weight loss	Cancer

Model learns:
Symptoms → Disease

Real-Life Examples:

- Email spam detection (Spam / Not Spam)
- House price prediction
- Student marks prediction
- Face recognition
- Credit card fraud detection
- Medical diagnosis (very important in biotech)
- Loan approval systems

Types of Supervised Learning:

- **Classification** → Output is category
(Spam / Not Spam)
- **Regression** → Output is number
(Price prediction)

2. Unsupervised Learning

Unsupervised Learning is when the model learns from **unlabeled data**.

No correct answers are given

Simple Idea:

“Learning without a teacher”

Model finds patterns on its own.

Example (Biotech):

Gene expression analysis

- Data: thousands of genes
- No labels
Model groups similar genes automatically

Real-Life Examples:

- Customer segmentation (grouping customers)
- Market basket analysis (products bought together)
- Social network analysis
- Image clustering
- Anomaly detection (fraud detection)
- Grouping patients with similar symptoms
- Recommendation systems (initial grouping)

Common Techniques:

- Clustering (K-means)
- Dimensionality reduction

3. Reinforcement Learning

Reinforcement Learning is when a model learns by **interacting with the environment** using **rewards and penalties**.

Simple Idea:

“Learning from experience”

Like trial and error.

Example:

Drug discovery simulation

- Model tries different drug combinations
- Gets reward if effective
- Penalty if not

Over time → learns best combination

Real-Life Examples:

- Self-driving cars 🚗
- Game playing AI (Chess, PUBG bots)
- Robotics 🤖
- Google Maps route optimization
- Stock trading bots
- Personalized recommendations
- Smart energy systems

Quick Comparison

Feature	Supervised	Unsupervised	Reinforcement
Data	Labeled	Unlabeled	No dataset (interaction)
Learning	With teacher	Without teacher	With reward
Example	Disease prediction	Customer grouping	Self-driving car

More Examples (For Broad Knowledge)

So that a student gets wide understanding

Supervised Learning Examples:

- Speech recognition
- Handwriting recognition
- Email classification
- Weather prediction
- Salary prediction

Unsupervised Learning Examples:

- News article grouping
- Topic modeling
- Image segmentation
- Customer behavior analysis
- Fraud pattern detection

Reinforcement Learning Examples:

- YouTube recommendations improving over time
- Chatbots learning from interaction
- Game AI (like chess engines)
- Robot learning to walk
- Ad placement optimization

Core ML Concepts

1. Input (Features)

Input (also called Features) are the data given to the model to learn from.

Represented as **X**

Simple Meaning:

“What we give to the model”

Example:

For predicting marks:

Hours Studied	Sleep Hours
2	6

Inputs = Hours Studied, Sleep Hours

2. Output (Target)

Output (also called Target or Label) is what we want the model to predict.

Represented as y

Simple Meaning:

“What we want from the model”

Example:

Hours Studied	Marks
2	30

Output = Marks

3. Model

A **model** is a mathematical function that learns the relationship between input and output.

Simple Meaning:

“The brain of the system”

Example:

Model learns:

Marks = $10 \times$ Hours Studied

So if:

Hours = 5 $\rightarrow$ Marks = 50

4. Training

Training is the process where the model learns from data.

Simple Meaning:

“Teaching the model”

Example:

You give data:

Hours	Marks
1	10
2	20
3	30

Model learns pattern:
 $\text{Marks} = 10 \times \text{Hours}$

5. Testing

Testing is checking how well the model works on new, unseen data.

Simple Meaning:

“Exam for the model”

Example:

Model learned from 1,2,3 hours

Now test:
Hours = 4

Expected:
40

If model predicts correctly → good model

6. Prediction

Prediction is the output given by the trained model for new input.

Simple Meaning:

“Final answer given by model”

Example:

Input:
Hours = 6

Prediction:
Marks = 60

Full Flow (Very Important)

This is how ML works step-by-step:

1. Give **Input (X)**
2. Give **Output (y)**
3. Train the **Model**
4. Test the model
5. Use it for **Prediction**

Complete Example (Scikit-learn Style)

Let’s connect everything with Scikit-learn

```
from sklearn.linear_model import LinearRegression

# Input (X)
X = [[1], [2], [3], [4]]

# Output (y)
y = [10, 20, 30, 40]

# Model
model = LinearRegression()

# Training
model.fit(X, y)

# Prediction
print(model.predict([[5]])) # Output: 50
```

Mapping Concepts to Code

Concept	Code Example
Input (X)	[[1], [2], [3]]
Output (y)	[10, 20, 30]
Model	LinearRegression()
Training	model.fit(X, y)
Testing	model.predict(new data)
Prediction	Final output